

Linear Algebra Refresher

A Visual & Interactive Refresher for Machine Learning and Data Mining

Vectors • Matrices • Geometry • Transformations • Eigenvectors • PCA • SVD

[Why LA?](#)[Vectors](#)[Dot Product](#)[Norms](#)[Distance](#)[Matrices](#)[Multiplication](#)[Systems](#)[Transformations](#)[Eigen](#)[PCA](#)[SVD](#)[Cheat Sheet](#)

1. Why Linear Algebra in Machine Learning?

Suppose we have 100,000 customers and 20 attributes for each customer. How should we represent this data?

Customer	Age	Income	Purchases	Visits
C1	25	40	5	12
C2	41	75	12	20
C3	32	55	8	15

One Customer → Vector

$$\mathbf{x} = \begin{bmatrix} 25 \\ 40 \\ 5 \\ 12 \end{bmatrix}$$

Dataset → Matrix

$$X = \begin{bmatrix} 25 & 40 & 5 & 12 \\ 41 & 75 & 12 & 20 \\ 32 & 55 & 8 & 15 \end{bmatrix}$$

Data



Vectors



Matrices



ML Algorithms

Key message: Machine learning is, to a large extent, mathematics operating on vectors and matrices.

2. Scalars, Vectors and Matrices

Scalar

$$x = 5$$

A single number.

Examples: age, temperature, learning rate.

Vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

An ordered collection of numbers.

Matrix

$$X \in \mathbb{R}^{m \times n}$$

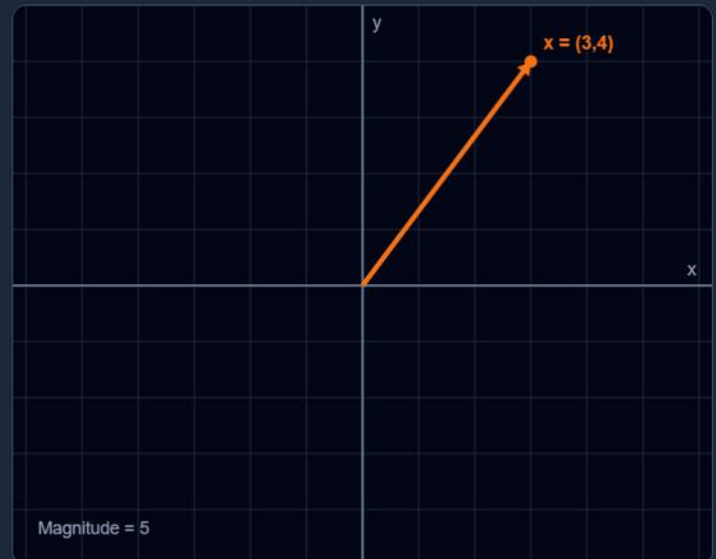
A rectangular collection of numbers.

Vector as a Geometric Object

$$\mathbf{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\|\mathbf{x}\| = \sqrt{3^2 + 4^2} = 5$$

A vector has both **direction** and **magnitude**.



Ask students: If a dataset has 50 features, what is the dimension of each observation vector? **Answer: 50.**

3. Vector Addition

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$\mathbf{a} + \mathbf{b} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Addition is performed component by component.

Vectors must have the same dimensions to be added.



4. Scalar Multiplication

$$\mathbf{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$3\mathbf{x} = \begin{bmatrix} 6 \\ 9 \end{bmatrix}$$

Multiplication by a scalar changes the length of a vector.

Important in:

- Gradient descent
- Optimization
- Normalization
- Parameter updates



5. Dot Product

One of the most important operations in machine learning.

$$\mathbf{x}^T \mathbf{w} = x_1 w_1 + x_2 w_2 + \cdots + x_n w_n$$

Interactive Vectors



$$x^T w = (3.00)(1.00) + (2.00)(4.00) = \boxed{11.00}$$

$$\theta = 42.27^\circ$$

Machine Learning Connection

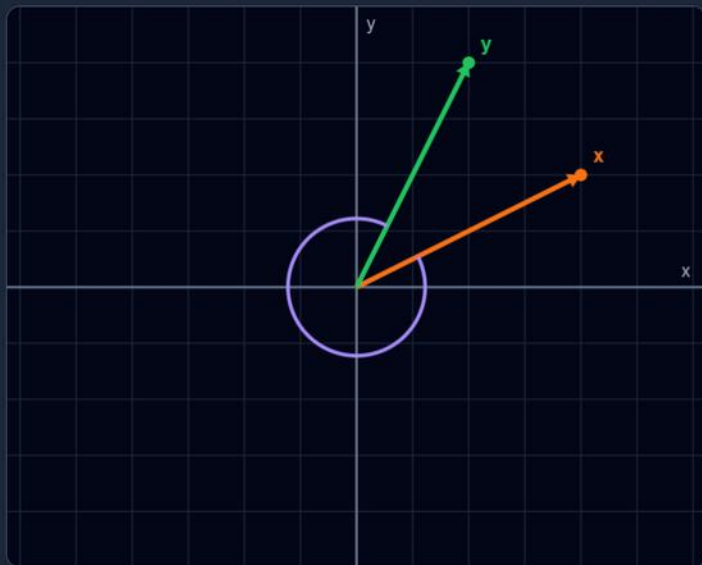
$$\hat{y} = \mathbf{w}^T \mathbf{x} + b$$

A linear model is essentially a weighted sum of the features.

6. Dot Product and Angle

$$\mathbf{x}^T \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta$$

$$\cos \theta = \frac{\mathbf{x}^T \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$



Cosine Similarity

$$\text{cosine}(x, y) = \frac{x^T y}{\|x\| \|y\|}$$

Useful in:

- Document similarity
- Text mining
- Information retrieval
- Recommendation
- Embeddings

If two vectors point in nearly the same direction:

$$\cos \theta \approx 1$$

they are highly similar in direction.

7. Vector Norms

A norm measures the size or magnitude of a vector.

L2 / Euclidean Norm

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

For

$$\mathbf{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\|\mathbf{x}\|_2 = 5$$

L1 Norm

$$\|\mathbf{x}\|_1 = \sum_i |x_i|$$

Example:

$$\left\| \begin{bmatrix} -2 \\ 3 \\ -4 \end{bmatrix} \right\|_1 = 9$$

ML Connection

$$L_1 : \lambda \|\mathbf{w}\|_1$$

$$L_2 : \lambda \|\mathbf{w}\|_2^2$$

Norms appear in regularization, optimization, distance calculations, and sparse modeling.

8. Distance Between Data Points

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2$$

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$\mathbf{x} - \mathbf{y} = \begin{bmatrix} -3 \\ -4 \end{bmatrix}$$

$$d(\mathbf{x}, \mathbf{y}) = 5$$



Data Mining

- K-means
- K-nearest neighbors
- Hierarchical clustering
- Anomaly detection

9. Matrices

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

This is a 2×3 matrix.

- 2 rows
- 3 columns

$$X \in \mathbb{R}^{m \times n}$$

If we have m observations and n features:

$$X \in \mathbb{R}^{m \times n}$$

Example:

$$X \in \mathbb{R}^{1000 \times 20}$$

1,000 observations and 20 features.

10. Matrix Addition

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

Matrices must have the same dimensions to be added.

11. Matrix Multiplication

$$(m \times n)(n \times p) = (m \times p)$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$



Important: Matrix multiplication is generally not commutative.

$$AB \neq BA$$

12. Matrix × Vector

$$W = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$W\mathbf{x} = \begin{bmatrix} w_{11}x_1 + w_{12}x_2 \\ w_{21}x_1 + w_{22}x_2 \end{bmatrix}$$

Neural Network Connection

$$\mathbf{z} = W\mathbf{x} + \mathbf{b}$$

A matrix transforms the input vector and the bias shifts the result.

13. Transpose

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$(AB)^T = B^T A^T$$

14. Identity Matrix and Inverse

Identity

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AI = IA = A$$

Inverse

$$AA^{-1} = A^{-1}A = I$$

Not every matrix has an inverse.

In machine learning, the inverse/pseudoinverse appears in solving linear systems and least-squares problems.

15. Systems of Linear Equations

$$2x + y = 5$$

$$x - y = 1$$

Write the system as:

$$A\mathbf{x} = \mathbf{b}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

ML connection: Many machine learning problems can ultimately be expressed in terms of systems of equations. For example:

$$X^T X w = X^T y$$

16. Linear Transformations

$$\mathbf{y} = A\mathbf{x}$$

Think of a matrix as an operator that transforms vectors.

Scaling

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Reflection

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Projection

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Interactive Transformation

Matrix A

a 2

b 0

